

ANKITA ARUN NATH

Data Engineer (Python, ETLA, ML) | Full-Stack Developer (React, Django/Flask)

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🔗 [ML based traffic scheduling techniques for intelligent transportation system \(Wiley Research\)](#)

📄 Student Visa (Germany) – Eligible to work 20 hrs/week

🐙 <https://ankita259.github.io/>



👤 PROFILE

Software Engineer with 5+ years of experience, including hands-on experience in full-stack web development alongside data pipelines and ETL processes. Skilled in Python, SQL, and building data-driven applications, including data ingestion using APIs and web scraping. With a strong foundation in React and Django/Flask, I bring an end-to-end perspective to problem-solving, connecting data processing with application development. Currently pursuing a Master's in Artificial Intelligence in Germany, with a focus on industrial applications of data and machine learning.

🧠 SKILLS

Programming Languages: Python (OOP, data processing), Html/Css/Javascript, GraphQL, Shellscript

Data Engineering: ETL pipelines, data ingestion, transformation, modeling | Web scraping, REST & GraphQL APIs | JSON, XML, SQL | Pandas, NumPy, TensorFlow (basic)

Libraries & Framework: Django, Flask, React, Vue.js, Node.js | Celery, Graphene | Pytest, Jest

Database: Elasticsearch, My SQL

Cloud and DevOps: AWS (Lambda, S3, RDS), Docker, Jenkins, CI/CD pipelines

Monitoring and Visualization: Grafana, Kibana, ELK Stack

Other tools and technologies: OpenStreetMap, SUMO (simulation tools) | Git/GitHub, Jira, Postman

📁 WORK EXPERIENCE

Data Engineer, [Data Axle India](#) 09/2025 – 03/2026 | Pune, India

- Built and maintained **ETL pipelines** to ingest data from web scraping, APIs, databases, and Excel sources
- Developed scalable **web scraping solutions (BeautifulSoup)** for structured and unstructured data extraction
- Transformed heterogeneous data into a **clean, normalized internal database** for downstream applications
- Contributed to a **subscription-based data platform** serving 40,000+ organizations across North America
- Implemented backend services using **Python (Django)** and frontend features with **React/Vue, GraphQL**
- Optimized **Celery workflows**, improving performance and reliability in distributed systems
- Resolved **production issues** and improved system stability
- Contributed to a **modular monorepo** with BDD-driven documentation
- **Tech Stack:** Python, Django, GraphQL, Web scraping & parsing (BeautifulSoup), ETL pipeline, Data Modeling, ML Algorithms, Celery, AWS, PLSQL, Javascript, React.js, Vue.js, Nx, Poetry, Industry 4.0, SOLID

Data Engineer, [Zensar Technologies](#) 01/2025 – 09/2025 | Pune, India

- Built **Python-based scraping pipelines** (UI + APIs) for 3 auction platforms, handling CAPTCHA-protected systems
- Processed **20,000+ records across 200+ sales**, extracting structured and unstructured data
- Designed **robust scraping logic** (BeautifulSoup, requests, regex) with fallback strategies
- Implemented **ML-based data validation** and automated **ETL pipelines** for reliable data ingestion into SQL
- Leveraged **AWS (Lambda, S3, RDS)** for scalable data processing and storage
- Developed a **fallback admin portal (Flask, React, Elasticsearch)** to ensure data availability
- Set up **CI/CD (Azure DevOps)** and **Dockerized deployments** for stable operations
- **Tech Stack:** Python, JavaScript, Flask, ReactJS, MYSQL, Elasticsearch, Git/Github, VS code, ML/AI, numpy, tensorflow, AWS/Azure, Jenkins

Senior Software Engineer, [Synopsis India Ltd.](#) 11/2022 – 01/2025 | Bengaluru, India

- Developed internal tools using **Python, JavaScript, and React**, improving productivity for 5,000+ users

- Automated workflows using **Python and Shell (Jenkins, Jira, Perforce APIs)**
- Ensured **code quality and test coverage** with Pytest and Jest, following CI/CD practices
- Provided **production support and monitoring** using Grafana and ELK stack
- Contributed to clean, maintainable codebases with **BDD-driven documentation**
- **Tech Stack:** Html/Css/Javascript, React.js, Redux, Thunk, Jest, Python, Flask, Cgi, Elasticsearch (ELK), MySQL, Grafana, Jenkins, Helixswarm, Perforce/Github, Jira, Postman

Software Engineer, [Tata Consultancy Services](#) 

10/2020 – 11/2022 | Bengaluru, India

- Developed a **user-facing complaint dashboard** using React and Flask (ARIA-compliant)
- Built and maintained **full-stack features** for a multinational retail client
- Extracted and processed data from **Snowflake databases** for application use
- Integrated **Java Spring Boot APIs** with frontend systems
- Managed **CI/CD and deployments** using Azure Kubernetes and Jenkins
- **Tech stack:** ARIA, Html/Css/Javascript, React.js, Redux, Thunk, Python, Flask, MySQL, Jenkins, Bitbucket, Sonarqube, Github, Jira, Postman, Kubernetes

EDUCATION

M.Sc. Artificial Intelligence for Industrial Application,

03/2026 | Germany

[Ostbayerische Technische Hochschule Amberg-Weiden](#) 

- Key Focus Areas: AI in Industry, Process Optimization, Machine-Sensor Data Handling

B. TECH (COMPUTER SCIENCE), [Vishwakarma Govt. Engg. College](#) 

08/2016 – 08/2020 | India

- GPA:1,7 and CGPA: 8.59/10

CLASS 12TH (H.S.C), [Swami Vivekananda English](#) - Medium school

05/2014 – 05/2016

- Score: 90.33%

OTHER PROJECTS

ML Based Traffic Police Scheduling

07/2019 – 04/2021

- Created a routine for real-time scheduling and dynamic deployment of traffic police based on machine learning and current traffic density.
- Used OpenStreetMap and SUMO simulator for traffic data simulation.
- Developed a cross-platform routine with OpenCV, TensorFlow, and Ionic SDK.

Vision Compass

10/2018

- Developed an Android app for the visually impaired that converts text from video to audio. Utilized OpenCV for image processing, Tesseract OCR, and Android Studio.

Hospital Management System

03/2018

- Created a tool to manage patient information, treatments, staff, and pharmacy details, ensuring efficient hospital operations.
- Tech stack: Html/Css/Javascript, Bootstrap, Flask, SQLite, JS, Python

LANGUAGES

English: IELTS Band C1

Deutsch/German: A2

COUSES & CERTIFICATES

[Deep Learning Specialization](#)  , Coursera

[Structuring ML Projects](#)  , Coursera

[React - The Complete Guide](#)  , Udemy

[Web App Technology\(Django\)](#)  , Coursera

[Programming for Everybody](#)  , Coursera

[Python Data Structures](#)  , Coursera

INTERESTS

- Art (sketching, painting), Travel, Music